Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

PROJECT TITLE

Algorithmic Trading Strategy Optimization via Deep RL

The domain of the Project:
Deep Reinforcement Learning

COURSE NAME:
AIML

Team Mentor (and their designation):
Mr. Gaurav Patel (Data Engineer)

Team Member:
Mr. Manoj Sundar P.

Period of the project
June 2026 to July 2026

Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled "Algorithmic Trading Strategy Optimization via Deep RL" has been mentored by Mr. Gaurav Patel, organised by SURE Trust, from June 2026 to July 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:
Signature: *Manoj Sundar.P*

Mentor's Name:
Mr. Gaurav Patel
Designation & Company Name:
Data Engineer, Celebal Technologies

Director's Name:
Prof. Radhakumari
Designation:
Executive Director & Founder, SURE Trust

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Executive Summary

This project develops and deploys an advanced Algorithmic Trading Strategy using Deep Reinforcement Learning (DRL) to address the limitations of static parametric models. By engineering features that represent volatility clustering, spoofing, and news shocks, the Continuous PPO (Proximal Policy Optimization) agent dynamically adapts to intraday regime shifts. The system utilizes a 3-State Hidden Markov Model (HMM) for regime detection and is fully deployed via a robust GitHub Actions CI/CD pipeline. The live paper-trading results and portfolio allocations are visualized in real-time through an interactive Streamlit dashboard connected to a Supabase Cloud Database.

Introduction

Background and context of the project

Traditional quantitative trading heavily relies on static parametric strategies. However, in modern algorithmic environments, markets are highly dynamic and adversarial.

Problem statement or goals of the project

Static parametric strategies cannot adapt to intraday regime shifts — spoofing, news shocks, and volatility clustering cause massive P&L drawdowns.

Scope and limitations of the project

The project's trading universe is strictly scoped to the 28 DJIA (Dow Jones Industrial Average) Equities plus a Cash asset. The initial baseline capital is simulated at $1,000,000.00. Execution is constrained to paper trading environments (via Alpaca) to safely evaluate performance without risking real capital.

Innovation component in the project

The primary innovation lies in marrying robust state representation (567-dimensional observation spaces capturing HMM market regimes and volume imbalances) with a Continuous PPO agent. Furthermore, the complete autonomous CI/CD pipeline ensures the model executes daily trades reliably, handles dependency resolution dynamically, and pushes analytics to a real-time web UI.

Project Objectives

- **Clearly defined objectives and goals of the project:** Engineer comprehensive features to represent volatility clustering, spoofing, and news shocks. Adapt a Gymnasium trading environment to consume these features and penalize massive drawdowns via Sharpe/Sortino ratios. Train an optimal Deep RL model (PPO).
- **Expected outcomes and deliverables:** A live execution pipeline ('trade_executor.py') integrating Alpaca API and Supabase Cloud, accompanied by a real-time Streamlit web dashboard to track portfolio trajectory, asset weights, and market regimes.

Methodology and Results

- **Methods/Technology used:** Proximal Policy Optimization (PPO) for continuous action-space trading, Hidden Markov Models (HMM) for 3-state market regime detection, Deep Reinforcement Learning for portfolio allocation optimization.
- **Tools/Software used:** Python, Stable-Baselines3, PyTorch, Gymnasium, Pandas, NumPy, Streamlit, Plotly, Supabase, GitHub Actions.
- **Data collection approach:** Fetches daily/intraday pricing for DJIA assets using the 'yfinance' API and calculates technical indicators.
- **Project Architecture:** The trained PPO agent evaluates a 567-dimensional state and outputs target portfolio weights. The execution script computes share allocations and logs trades to a Supabase Cloud Database. A GitHub Actions CI/CD pipeline triggers this daily, and a Streamlit dashboard visualizes the live data.
- **Final project working screenshots:**

Figure 1: Main Streamlit Dashboard View

- **Project GitHub Link:** `https://github.com/AKISIONOV/Reinforcement-Learning-Model-f`

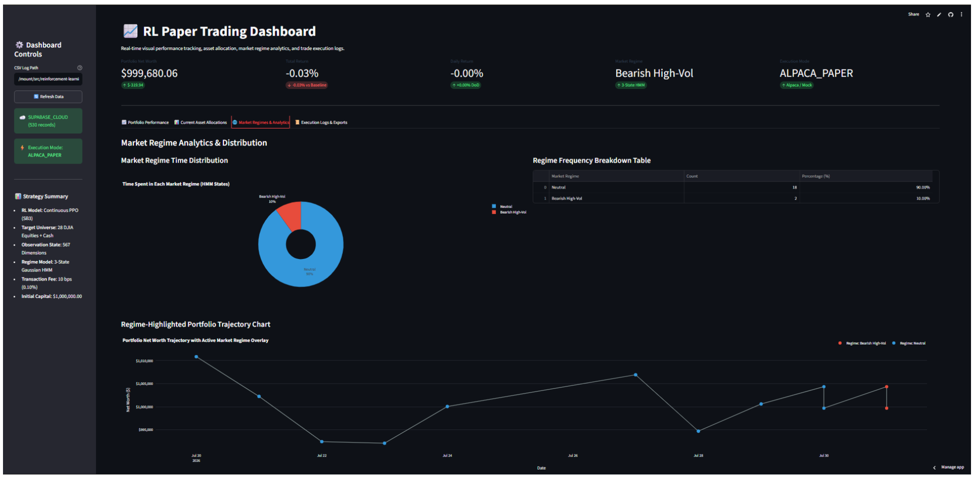

Figure 2: Market Regime Detection (HMM)

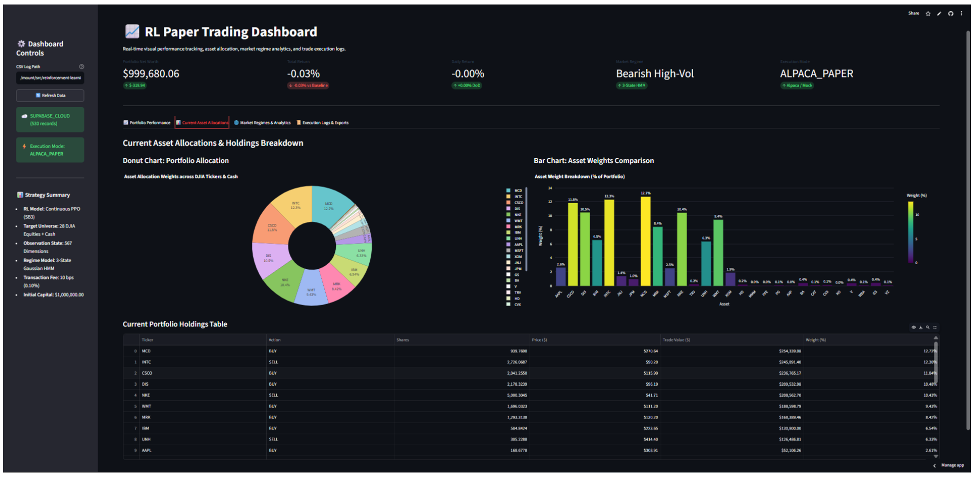

Figure 3: Live Portfolio Asset Allocation

Social / Industry relevance of the project

In the financial industry, capital preservation is as critical as capital generation. By engineering an AI agent capable of detecting market manipulation (spoofing) and avoiding sudden shocks, this project offers a highly robust risk-management framework. Such adaptive models provide immense value to quantitative hedge funds, proprietary trading firms, and retail algorithmic traders looking to protect their portfolios against flash crashes.

Learning and Reflection

- **Document my new learnings:** Gained profound hands-on experience in training deep reinforcement learning models for continuous control tasks using Stable-Baselines3. Mastered CI/CD automation using GitHub Actions and learned how to bridge backend data engineering with frontend visualization using Streamlit and Supabase.
- **Document my experience:** Tackling complex Python dependency conflicts (NumPy 1.x vs 2.x serialization) and utilizing autonomous multi-agent teams for debugging provided an incredible, real-world perspective on modern AI-assisted software development.

Conclusion and Future Scope

- **Recap objectives and achievements:** The project successfully demonstrated that Deep RL agents equipped with comprehensive market regime awareness can dynamically adapt to adverse market conditions, preventing massive drawdowns. The end-to-end deployment pipeline serves as a reliable framework.
- **Future scope of this project:** Expand the target universe beyond DJIA equities to include Cryptocurrencies and Forex markets. Incorporate live Level-2 (L2) Order Book data to detect spoofing at the microsecond level, and combine PPO with SAC (Soft Actor-Critic) to create a robust ensemble policy.